

Data Visualization & Interactive Dashboard

Project Report based on the supplied Power BI report package (PBIX). The report documents the dashboard structure, analytical design, measures, interactivity and recommended interpretation.

Platform Microsoft Power BI	Core Entity NFLX	Visuals 5 configured visuals
Time Intelligence Year → Quarter → Month → Day	Trend Measures 20-day & 50-day moving averages	Interactive Filters Date + metric selector

Prepared from the supplied project file • September 2026



Embedded visual background asset found in the Power BI package.

1. Executive Summary

This project is an interactive Power BI dashboard for exploring Netflix (NFLX) stock-market data. The supplied report definition contains a dedicated NFLX data entity, a DateTable for time-based analysis, a metric selector named AVG_METRIC, and a measures table containing 20-day and 50-day moving-average measures. The dashboard is designed to let a user examine stock-related activity over time and switch between trend measures through interactive controls.

The visual layer is intentionally compact: two line charts focus on time-series trends, one column chart focuses on volume-related activity, and two slicers provide user-driven filtering. The date hierarchy supports drill-down from year to quarter, month and day. A date range is configured through 3 June 2022 in the supplied report definition.

Project objectives

- Convert stock-market data into an interactive visual analytics experience.
- Provide time-based trend analysis using Power BI date hierarchies.
- Compare short- and medium-term moving-average measures.
- Allow users to filter the dashboard by date and analytical metric.
- Present volume-related activity alongside trend-oriented visuals.
- Create a reusable foundation for deeper financial-data exploration.

Project outcome

The result is a structured, interactive dashboard design that combines raw stock fields, time intelligence, calculated measures and slicer-driven exploration in a single Power BI report.

2. Project Architecture

NFLX STOCK DATA Close • Volume • related stock fields	DATA MODEL NFLX + DateTable + Measures	TIME INTELLIGENCE Year → Quarter → Month → Day
CALCULATED ANALYTICS 20-day moving average 50-day moving average	INTERACTION LAYER Date slicer AVG_METRIC selector	VISUAL OUTPUT Line charts + volume chart

Logical flow

Source / dataset	Transformation & model	Analytics	Dashboard
NFLX stock fields	DateTable + relationships	Moving averages	Time-series visuals
Close / Volume	Date hierarchy	20-day / 50-day	Volume + filters

Architecture note: The report definition is configured as a drillthrough page with Year as the drillthrough filter. This supports a focused year-level analytical workflow.

3. Data Model & Analytical Components

Component	Observed role in report	Fields / measures visible in definition
NFLX	Primary stock-data entity	Close; Volume
DateTable	Time dimension and filtering	Date; Date Hierarchy: Year, Quarter, Month, Day
Measuress	Calculated analytical measures	20_day_moving; 50_day_moving
AVG_METRIC	Interactive metric selector / field parameter	AVG_METRIC

Moving-average methodology

A moving average smooths a time series by calculating an average over a rolling window. In this project, the report exposes two windows: 20 days and 50 days. Conceptually, for a series x , a k -day moving average at time t can be represented as:

$$MA_k(t) = (x_{t-k+1} + \dots + x_t) / k$$

The 20-day window provides a shorter-term smoothing view, while the 50-day window represents a longer smoothing horizon. The report definition uses these measures directly in line-chart visuals.

4. Dashboard Visuals & Interactivity

Visual	Type	Purpose	Interaction
Stock trend	Line chart	Displays Close together with 20-day and 50-day moving-average measures over the Date hierarchy.	Responds to date and metric filtering.
Moving-average comparison	Line chart	Focuses directly on the 20-day and 50-day moving-average measures.	Supports time-level exploration.
Volume activity	Column chart	Uses the NFLX Volume field as a count-based visual across the Date hierarchy.	Supports time drill-down.
Metric selector	Slicer	Exposes the AVG_METRIC field parameter used by the moving-average visual.	User selects the metric configuration.
Date filter	Slicer	Provides a between-date selection using DataTable.Date.	Filters the analytical view by date.

Time hierarchy and drill-down

The visuals are configured with a Date Hierarchy containing Year, Quarter, Month and Day. Year is active at the top level in the visual definitions, while the lower levels are available for drill-down. This enables the same dashboard to support high-level trend review and more granular time exploration.

Configured date boundary

The supplied date slicer is configured in Between mode with an end date of 3 June 2022, and the page filter contains a corresponding boundary at 4 June 2022. This describes the configuration found in the supplied report; it should not be interpreted as the complete underlying dataset range.

5. Analytical Interpretation

What the dashboard is designed to reveal

Analytical question	Dashboard evidence	How to interpret it
How does stock activity change over time?	Time-series line chart using Close and moving averages.	Look for direction and sustained movement rather than isolated points.
How does short-term movement differ from longer-term movement?	20-day vs 50-day moving-average measures.	Compare the relative position and slope of the two smoothed series.
When is trading activity higher or lower?	Volume-related column chart.	Compare the height of columns across the selected time period.
Does the pattern change at different time scales?	Year / Quarter / Month / Day hierarchy.	Drill from broad periods into finer periods to inspect local changes.
How does the selected date range affect the story?	Between-date slicer.	Use narrower and wider windows to separate long-term patterns from short-term variation.

Important interpretation note: The supplied PBIX definition exposes the visual logic and measures, but the compressed Power BI DataModel was not decoded into a row-level table in this report. Therefore, this report documents the analytical design and intended reading of the dashboard rather than inventing numerical findings that are not directly visible in the supplied report definition.

6. Technical Implementation

Area	Implementation observed
BI platform	Microsoft Power BI report definition (PBIX package).
Report canvas	1920 × 1080 page configured with FitToPage display.
Themes	Fluent2-CY26SU08 base theme with the Innovate custom theme.
Time intelligence	DateTable with Date Hierarchy levels: Year, Quarter, Month, Day.
Calculated measures	20_day_moving and 50_day_moving measures in Measures.
Field parameter	AVG_METRIC selector used by the moving-average line chart.
Interactivity	Date slicer, metric slicer, drill-down hierarchy and drillthrough by Year.
Visual layer	2 line charts, 1 column chart and 2 slicers.

Implementation workflow

- **1. Load:** Bring the stock dataset into Power BI and expose the NFLX entity.
- **2. Model:** Add a dedicated DateTable and use a date hierarchy for time-based analysis.
- **3. Measure:** Create rolling-window measures for 20-day and 50-day averages.
- **4. Parameterize:** Provide AVG_METRIC as a field-parameter selector for flexible trend analysis.
- **5. Visualize:** Combine line and column charts with slicers.
- **6. Interact:** Enable drill-down, date filtering and Year-based drillthrough.

7. Future Enhancements

Enhancement	Purpose / benefit
Add KPI cards	Display latest Close, period change, selected-period high/low and volume summaries.
Improve volume semantics	Use the actual Sum of Volume rather than a count-based aggregation if the business requirement is trading-volume magnitude.
Add return metrics	Introduce daily/weekly returns and percentage change to complement price-level charts.
Add volatility analysis	Calculate rolling standard deviation or another clearly defined volatility measure.
Enhance navigation	Create dedicated overview and drillthrough pages for year, quarter and month analysis.
Add data-quality checks	Validate missing dates, duplicate records, null values and market-calendar gaps.
Document source provenance	Record the original dataset source, refresh date, license and update process.
Enable refresh workflow	Connect the model to a refreshable source so the dashboard can be maintained beyond the current configured period.

Conclusion

The supplied Power BI project demonstrates a focused stock-market visualization workflow built around NFLX data. Its strongest structural features are the dedicated time dimension, rolling-average measures, interactive metric selection, date filtering and drillthrough configuration. With additional KPI, return, volatility, data-quality and refresh capabilities, the project can be developed into a more comprehensive financial analytics dashboard.

Appendix: evidence captured from the PBIX definition

- Report contains one active page named "Page 1".
- The page is configured as a drillthrough page using DataTable.Year.
- Five visuals are defined: two line charts, one column chart and two slicers.
- The NFLX entity is referenced through Close and Volume fields.
- The Measures table exposes 20_day_moving and 50_day_moving.
- The AVG_METRIC entity is used as a field parameter in the moving-average visual.
- The Date slicer is configured in Between mode with an end date of 3 June 2022.